Constraint-Gated Waveform Design for Low-Excitation Ion Transport

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Abstract

We present a minimal, reproducible pipeline for designing trapped-ion transport waveforms and directly evaluating their dynamical consequences. Using a one-dimensional model, we show that smooth minimum-jerk transport trajectories suppress motional excitation by multiple orders of magnitude compared to linear interpolation, and we quantify clear tradeoffs between transport speed, waveform smoothness, and residual excitation. The framework is intentionally simple so that these relationships are transparent and directly inspectable, providing a practical bridge between waveform design and physical performance.

1 Introduction

Segmented trapped-ion architectures require reliable ion transport between trap zones. Transport waveforms must move ions quickly while limiting residual motional excitation. This work develops a small, reproducible simulation pipeline that directly connects waveform design to excitation outcomes.

This work focuses on making these tradeoffs directly visible through a minimal computational model, rather than optimizing a specific hardware implementation.

2 Related Work

Ion transport in segmented traps has been studied extensively in both experimental and theoretical contexts. Early work demonstrated reliable transport and splitting operations in RF traps [2], while later experiments achieved fast transport with low excitation using optimized control protocols [1].

From a theoretical perspective, smooth trajectory design has been shown to suppress excitation during transport [3], motivating the use of minimum-jerk and related interpolation schemes.

The present work complements this literature by providing a minimal, reproducible pipeline in which these effects are directly visible and quantifiable.

3 Pipeline

The computational pipeline is:

electrode basis $\rightarrow$ potential well $\rightarrow$ transport path $\rightarrow$ voltage waveform $\rightarrow$ ion motion $\rightarrow$ excitation metric.

4 Transport Model

Ion motion is modeled as a particle in a moving harmonic well:

$$m\ddot{x}(t) = -m\omega^2[x(t) - x_c(t)],$$

where $x_c(t)$ is the waveform-controlled well center.

5 Waveform Design

Smooth transport paths are generated using minimum-jerk interpolation:

$$s(u) = 10u^3 - 15u^4 + 6u^5.$$

This ensures zero velocity and acceleration at endpoints, reducing excitation.

6 Results

6.1 Transport Trajectory

The ion trajectory closely follows the target transport path under minimum-jerk control, indicating low excitation during motion.

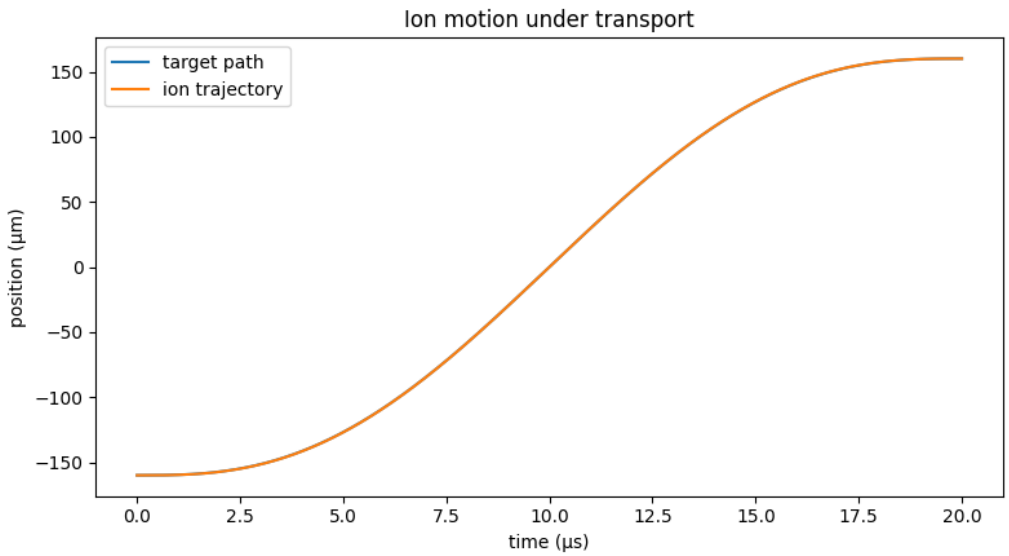

Figure 1: Ion trajectory closely follows the target transport path under minimum-jerk control.

6.2 Path Comparison

Minimum-jerk transport significantly suppresses oscillatory residual motion compared to linear interpolation, which strongly excites the trap.

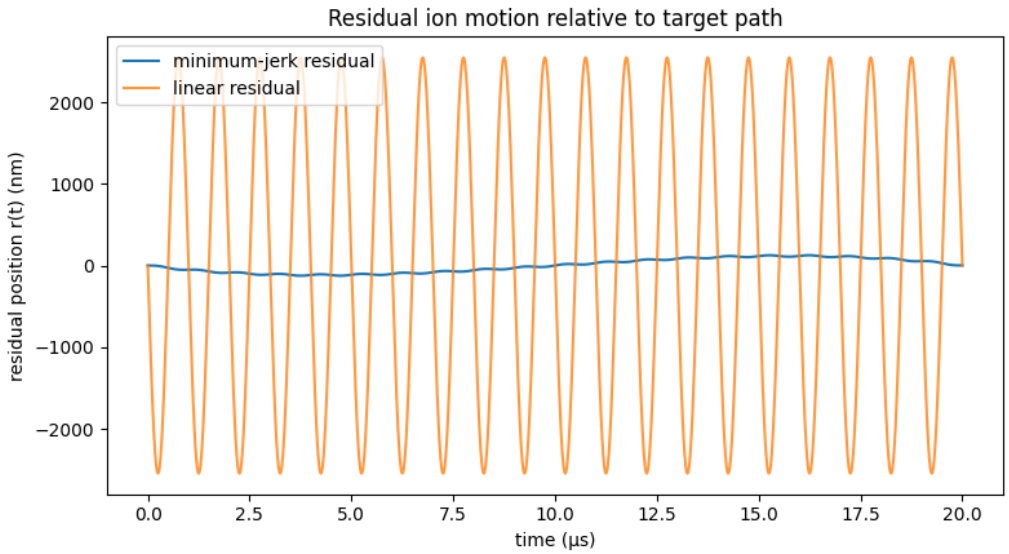

Figure 2: Minimum-jerk transport suppresses oscillatory excitation relative to linear interpolation.

6.3 Excitation Scaling

Residual excitation decreases systematically with increasing transport duration, demonstrating a clear tradeoff between speed and excitation.

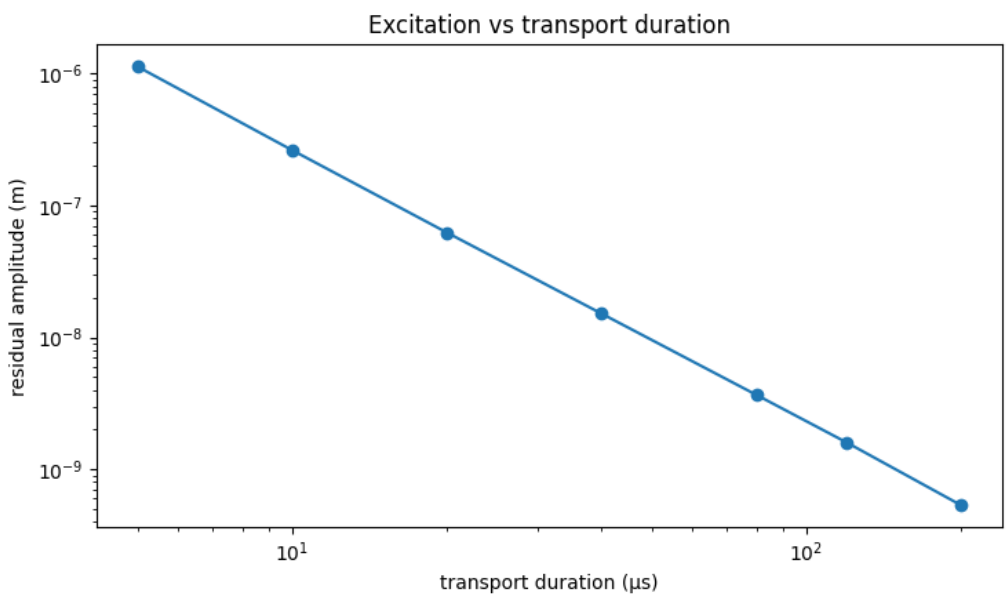

Figure 3: Residual excitation decreases systematically with increasing transport duration.

6.4 Frequency-Domain Analysis

Frequency-domain analysis shows that linear transport excites the trap near the secular frequency, while minimum-jerk trajectories suppress resonant excitation.

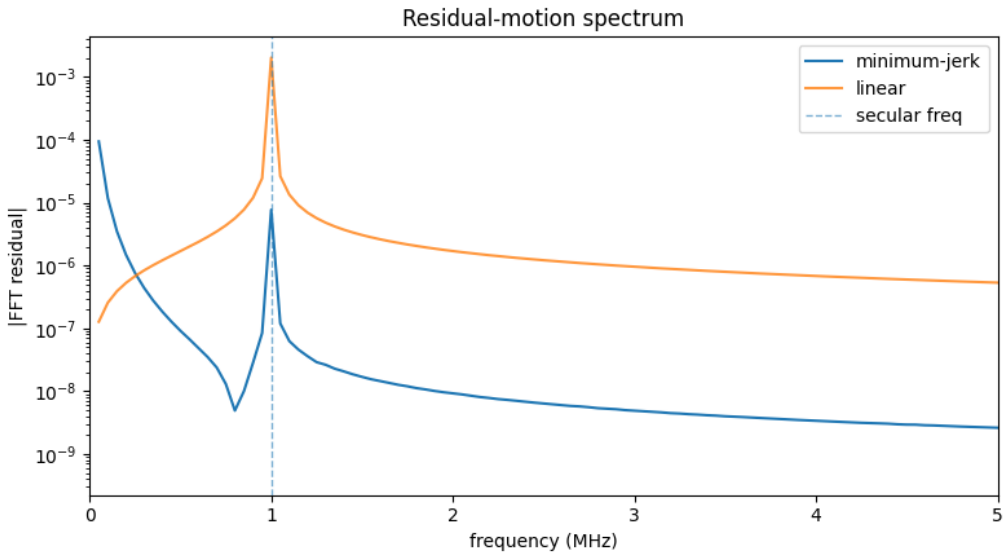

Figure 4: Residual-motion spectrum showing suppression of excitation at the trap secular frequency.

7 Discussion

The key result of this work is that excitation behavior emerges clearly from trajectory smoothness alone, without requiring complex control optimization.

Smooth transport paths suppress resonant excitation by minimizing dynamical forcing. The results demonstrate:

- Orders-of-magnitude reduction in excitation for minimum-jerk paths
- Clear scaling laws linking transport speed and excitation
- Practical control constraints arising from waveform smoothness

8 Conclusion

We presented a compact, interpretable pipeline for ion transport waveform design. The framework directly connects control synthesis to physical outcomes and highlights fundamental tradeoffs between speed and excitation. This approach provides a foundation for more advanced optimal control and experimental implementation.

Code Availability

All code and figures are generated from a small, reproducible pipeline available at:
github.com/thinkthoughts/ion-transport-waveform-pipeline

References

[1] R. Bowler, J. P. Gaebler, Y. Lin, et al. Coherent diabatic ion transport and separation in a multizone trap array. *Physical Review Letters*, 2012.

[2] M. A. Rowe, D. Kielpinski, V. Meyer, et al. Transport of quantum states and separation of ions in a dual rf ion trap. *Quantum Information and Computation*, 2002.

[3] E. Torrontegui, S. Ibáñez, S. Martínez-Garaot, et al. Fast atomic transport without vibrational heating. *Physical Review A*, 2011.